

CHARALAMBOS HADJIPANAYI

Postdoctoral Researcher

Biomedical Signal Processing & Digital Health

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SOCIAL MEDIA

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RESEARCH PROFILES

ORCID iD: 0000-0002-2624-6178

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NATIONALITY

Cypriot

British

LANGUAGES

English

Greek

EDUCATION

09/2020 – 02/2025 **Ph.D., Electrical and Electronic Engineering**

Imperial College London

- **Thesis:** "Remote Gait Analysis for Assisted Living using Radar-on-chip Technology". Developed and validated contactless radar-based methods for extracting clinically relevant spatiotemporal gait parameters, with validation in healthy participants and neurological populations.
- **Award:** Best Live Demonstration, IEEE BioCAS 2024

10/2016 – 06/2020 **M.Eng., Bioengineering (First Class, 80.91%)**

Imperial College London

- **Thesis:** "Cybathlon 2020 Powered Exoskeleton Race". Developed closed-loop sensory feedback for lower-limb exoskeleton control using plantar-pressure measurements.
- **Awards:** Dean's list 2019 (80.1%), 2018 (86.2%), 2017 (84.3%)

RESEARCH EXPERIENCE

02/2025 – Present **Research Associate, Radar Signal Processing**

Imperial College London, UK DRI Care Research & Technology Centre / NGNI Lab

- Radar-based behavioural and physiological sensing for healthcare.

07/2024 – 02/2025 **Research Assistant, Radar Signal Processing**

Imperial College London, UK DRI Care Research & Technology Centre / NGNI Lab

ENGINEERING / WORK EXPERIENCE

07/2019 – 09/2019 **Summer Intern**

Dept. of Electrical and Computer Engineering, University of Cyprus

- Radar-based vital-sign extraction (ECG-validated) under the supervision of Prof. Constantinos Pitris.

07/2018 – 08/2018 **Assistant Technician**

Scientronics Ltd., Cyprus

- Medical device installation and maintenance.

06/2015 – 07/2016 **Electronic Telecommunication Operator**

Cyprus National Guard General Staff (compulsory military service)

- Operated electronic communications and managed the social media webpage of the National Guard General Staff.

TEACHING EXPERIENCE

02/2021 – 09/2024 **Graduate Teaching Assistant**

Department of Bioengineering, Imperial College London

- Supervised laboratories and assessed coursework and examinations.
- Head teaching assistant for "Digital Biosignal Processing" module in 2024.
- Attended and actively participated in "Learning and Teaching" and "Assessment and Feedback for Learning" workshops.

SKILLS

Programming Languages

- MATLAB (Expert)
- Python (Advanced)
- C / C++ (Intermediate)
- HTML / CSS (Intermediate)
- JavaScript (Beginner)
- VBA (Beginner)

Scientific & Technical Software

Modelling & Analysis

- Simulink, Wolfram Mathematica, Jupyter Notebooks, Unity

Motion Capture & Biomechanics

- OpenSim, Vicon Nexus

Documentation, Communication & Collaboration

- Scientific writing & documentation (LaTeX, Microsoft Word, Microsoft Excel)
- Presentation & poster design (Microsoft PowerPoint, Microsoft Publisher)
- Collaboration & project management (Git / GitHub, Microsoft Planner, Notion)
- Media & content creation (Shotcut, Inkscape, OBS)

Prototyping & System Development

Embedded & Rapid Prototyping

- Arduino, Raspberry Pi

Circuit Design & Simulation

- OrCAD, PSpice, Intel/Altera Quartus

CAD Design, Mechanical Prototyping & Fabrication

- SolidWorks, Autodesk Fusion 360, 3D Printing

PUBLICATIONS

2026

- **C. Hadjipanayi** et al., “Towards Radar-Agnostic Gait Analysis Across UWB and FMCW Systems,” in Proc. Int. Conf. IEEE EMBC, 2026.
- M. Yin, K. G. Ravindran, **C. Hadjipanayi**, et al., “Using Legacy Polysomnography Data to Train a Radar System to Quantify Sleep in Older Adults and People Living with Dementia,” IEEE Trans. Biomed. Eng., 2026.
- H. Tu, M. Yin, **C. Hadjipanayi**, et al., “Multi-Person Sleep Monitoring Using Ultra-Wideband Radar and Deep Learning,” in Proc. Int. Conf. IEEE EMBC, 2026.
- Z. Chen, **C. Hadjipanayi**, et al., “Towards Continuous Home Monitoring for Dementia: A Real-Time mmWave Radar Framework for Activity Classification and Tracking,” IEEE J. Biomed. Health Inform. (JBHI), 2026.

2025

- J. J. Yun, **C. Hadjipanayi**, et al., “Passive Sensing of Gait and Medication-Related Fluctuations in Parkinson’s Disease,” J. NeuroEng. Rehabil., 2025.
- M. Yin, **C. Hadjipanayi**, et al., “Unobtrusive Sleep Health Assessment Using Impulse Radar: A Pilot Study in Older People,” IEEE Trans. Biomed. Eng., 2025.
- A. Bannon, A. Rapeaux, O. Savolainen, **C. Hadjipanayi**, et al., “Enabling Low-Cost Ultra-High-Framerate WiFi-Networked UWB Radar Systems for Indoor Monitoring of Physiology and Activity,” in Proc. IEEE Int. Radar Conf. (RADAR), 2025.

2024

- **C. Hadjipanayi** et al., “Remote Gait Analysis Using Ultra-Wideband Radar Technology Based on Joint Range-Doppler-Time Representation,” IEEE Trans. Biomed. Eng., 2024.
- **C. Hadjipanayi** et al., “Capturing Gait Parameters During Asymmetric Overground Walking Using Ultra-Wideband Radars: A Preliminary Study,” in Proc. Int. Conf. IEEE EMBC, 2024.
- Z. Chen, **C. Hadjipanayi**, et al., “Live Demonstration: Real-Time Gait and Respiration Analysis Using Ultra-Wideband Radar,” in Proc. IEEE BioCAS, 2024.
- M. Yin, **C. Hadjipanayi**, et al., “Human Gait Identification Using UWB Radar Micro-Doppler Signature,” in Proc. Int. Conf. IEEE EMBC, 2024.

2023

- M. Crook-Rumsey, S. J. C. Daniels, S. Abulikemu, H. Lai, A. Rapeaux, **C. Hadjipanayi**, et al., “Multicohort Cross-Sectional Study of Cognitive and Behavioural Digital Biomarkers in Neurodegeneration: The Living Lab Study Protocol,” BMJ Open, 2023.

PATENT APPLICATION

Radar System and Associated Methods

- International patent application: WO2025/229322 (PCT/GB2025/050916); published 6 November 2025
- Priority application: GB 2406092.3 (filed 30 April 2024)
- Inventors (equal): T. Constandinou, N. Yogendran, A. Bannon, A. Rapeaux, **C. Hadjipanayi**, M. Yin